# Jules Couprie

*Aerodynamics & CFD Engineering Intern*

4 month internship| Available from May 2027

## Contact


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Portfolio - More detailed resume


## Hard skills

- CAD - 3DX / Catia V5 / Solidworks
- CFD - Ansys suite / Paraview
- MATLAB / Simulink
- Python (Pycharm)
- Microsoft Suite - Excel

## Soft skills

**Project Management:**
Structured engineering workflows, defined task milestones, managed project deadlines

**Team Leadership**
Coordinated cross-functional student teams to solve complex problems.

**Adaptability & rigour:**
Quick integration into a team, structured methodology and validation of results

## Hobbies

**Sports & Endurance**
- Finisher of the GR20
- Trail running - marathon runner
- Climbing
- 8 years skateboarding

**Media Production**
- Videography & Photography
- Video editing (Davinci resolve)

## Language

**French** - Native

**English**- Full professional proficiency
*TOEIC 960 / Cambridge assessment*

**German**- Limited working proficiency

## Profile

Engineering student in 4th year at ESTACA, specialising in aeronautical technologies. Experience in CFD, FSI simulation, dynamic meshing and aerodynamic optimisation. Seeking a 4-month internship in aerodynamics applied to high-performance sport.

## Experience

### Supervised project - F1 Front Wing (May 2026 – Present)

*CFD design and study of a Formula 1 front wing*
**Supervision:** ONERA, ESTACA
**Technical support:** Louis Guidet, Aerodynamicist at Alpine F1 Team

- Performance assessment through L/D ratio, total pressure coefficient and sidewash, using the k-ω SST turbulence model and y+ ≈ 1
- Parametric geometry in 3DEXPERIENCE and CFD simulations in Ansys Fluent to study the effect of parameters at each phase (2D single-element airfoil → 3D multi-element wing compliant with FIA regulations)
- Aero mapping, shape optimisation and reduced computational cost via interpolation and a surrogate model (kriging)
- Automation of the CAD-CFD-post-processing workflow in Python, including an airfoil plotting application developed with AI assistance

### ČVUT - Prague (June - August 2026)

*Aero Research Intern*

Design and simulation of a linear peristaltic pump:

- Two-way FSI simulation (ANSYS Mechanical, Fluent, System Coupling) of a tube compressed by anvils
- CAD/mesh preparation: parametric geometry in 3DEXPERIENCE
- Dynamic meshing and remeshing techniques to handle large structural deformations
- Performance assessment through flow rate, backflow and stagnation detection, using Python post-processing scripts

### HYCE - iDrogen (2025 - Present)

*Head of Communications & Embedded Systems Team Member*

Project to convert a race car to a hydrogen-fuelled combustion engine, aiming for the record in our category at the Nürburgring circuit.

- Led project communication: team management, technical content creation (website, videos), press and media relations
- Sized the hydrogen supply system in Simulink to ensure vehicle safety, in compliance with FIA regulations

### SD Aviation - La Roche-sur-Yon (July 2024)

*Flight operator*

- Managed flight planning and scheduling using Leon and ForeFlight software.
- Operational support: Managed an online marketplace and performed various technical and manual maintenance tasks. Compliance matrices and quality control

## Education

### ESTACA, Laval (2023- Present)

*Master's Degree in Aeronautical Engineering*

- Specialisation in mechanics and advanced aeronautical technologies
- UCAV Design project on CATIA (Strategic and technical analysis, aerodynamic design and dimensioning, CAD modeling and simulation)
- Courses: Mathematics, Fluid mechanics, Aerodynamics, System mechanics, SOLIDWORKS, CATIA/3DX, MATLAB, CFD (Ansys)